

# DANIEL NG

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## Education

### University of British Columbia

BASc in Electrical Engineering (*Co-op*, available May 2026)

Expected Graduation: May 2028

Vancouver, BC

## Experience

### Norsat

June 2025 – August 2025

Hardware Test Intern

Richmond, BC

- Characterized Ku-Band Block-Up Converters using Vector Network Analyzers, Signal Analyzers, oscilloscopes, and multimeters to ensure 100% compliance with hardware specifications.
- Optimized RF power module stability by precisely sweeping DAC control codes, significantly reducing variance in output power prior to final acceptance testing.
- Assembled and troubleshoot BUCs and LNBs at the PCBA level, performing microscopic SMT and THT soldering and inspection to maintain quality standards for MMIC chips.
- Ensured 100% yield across a production run of 50+ BUC units by identifying DC bias instabilities through schematic cross-referencing.

### UBC Formula Design Team

September 2025 – Present

Electrical & Firmware: Project Lead

Vancouver, BC

 [GitHub](#)

- Lead end-to-end development of a laser receiver, incorporating optomechanical shrouding with a custom 3.3 V CMOS analog front end and 2.4 GHz ESP-NOW communication between master-slave ESP32-S3 MCUs to calculate speed and lap times.
- Characterize baseline photodiode thermal dark current and TIA voltage noise to calibrate an adjustable hysteresis threshold via potentiometer, targeting a 20 dB SNR and  $BER < 10^{-7}$  to ensure high-speed data integrity.

## Projects

### Oven Reflow Controller | A51 Assembly, CV-8052, NEC Protocol, I<sup>2</sup>C

January 2026

 [GitHub](#)

- Architected a bare-metal thermal control system on a CV-8052 soft-core (DE10-Lite FPGA), implementing a deterministic FSM with adaptive PWM-based closed-loop control.
- Integrated I<sup>2</sup>C peripherals, NEC IR protocol decoding (timing-validated bitstream parsing), and UART telemetry within a non-blocking interrupt-driven architecture.
- Developed a Python-based calibration and validation framework (NumPy/Matplotlib/pySerial) to stream live ADC data, perform thermocouple linearization and cold-junction compensation, cross-validate against laboratory instrumentation, and inject corrected temperature feedback to the controller — achieving  $\pm 3^{\circ}\text{C}$  closed-loop accuracy across 25–240°C.

### FPGA VGA Driver | Embedded C, DE10-Lite FPGA, VGA

November 2025

 [GitHub](#)

- Developed a bare-metal VGA driver using Memory-Mapped I/O (MMIO) to interface with an FPGA pixel buffer, implementing efficient 2D-to-1D address mapping via bitwise shifting.
- Engineered a pixel-processing pipeline to convert 24-bit RGB values to the RGB565 color format, using bit-masking and packing to optimize memory bandwidth.
- Engineered an autonomous robot opponent with predictive collision logic, performing real-time memory reads of the VGA buffer to scan for obstacles and update movement vectors.

## Skills

**Digital Design/HDL/Low-Level:** SystemVerilog, RISC-V, A51 Assembly, FPGA

**Embedded Systems:** CAN, USB, SPI, UART, I<sup>2</sup>C, ESP32, 8051-family MCUs, C Embedded Firmware, Bare Metal

**Design Tools:** Quartus Prime, Questa/ModelSim, Altium Designer, CPULator

**Hardware & Lab:** VNA, Signal Analyzer, PCB Design and Debugging, SMT/THT Soldering, Oscilloscope, Multimeter

## Hobbies/Interests

Competitive Badminton • Chess • Reverse Engineering Hardware • Parallel Computing • Graphics Pipeline Architecture